



Report PhD thesis Bram Vancraeynest-De Cuiper

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ANALYSIS OF THE DISSERTATION

This thesis is situated in the field of quantum many body physics. More specifically, it concerns the study of symmetries, a subject which has undergone a revolution in the past decade. Symmetries have long been recognised to be one of the primary organising principles in physics, in particular in the classification of phases of matter. This classification was brought into question with the discovery of topological phases of matter, which seemingly defy the usual Landau paradigm of symmetry breaking. More recently however, it has been understood that the Landau paradigm survives provided that one generalises the notion of symmetry. These generalised symmetries need not act on the full system, and need not be described by groups. Instead, they are described by (higher) category theory, a branch of mathematics renowned for its high level of abstraction.

The work in this thesis focuses on the realisation of these aspects in quantum lattice models. In this context, the abstract mathematical structures take on a very concrete form in the language of tensor networks. These reveal the entanglement structures inherently present in these generalised symmetries, and essentially provide the lattice representation theory of (higher) fusion categories. The results in this thesis are at the cutting edge of this rapidly developing field, and present a significant milestone in our understanding of symmetries in the quantum many body problem. The thesis contains 4 chapters reviewing the relevant background, followed by 6 chapters containing the published works of the author. These chapters can be summarised as follows:

After an introduction, chapter 2 introduces tensor network notation, concepts, and relevant theory for the remainder of the work. This includes matrix product states (MPS), projected entangled pair states (PEPS), as well as their operator versions. This chapter also includes a discussion of the fundamental theorem of matrix product states, the workhorse of tensor network theory. The chapter concludes with an application of this theorem; tensor network-based classification of gapped phases in $(1+1)d$.

Chapter 3 delves into the theory of topological phases in $(2+1)d$, and the associated mathematical structures. After an exposition on fusion categories, there is a detailed explanation of

the Levin-Wen string-net condensation construction. These lattice models are renormalisation group fixed-point representatives of the most general type of non-chiral topological order in $(2+1)d$. Importantly, the role of symmetries is emphasised: string-net states are nothing but symmetry breaking states of a 1-form symmetry described by a modular fusion category, and the quasiparticle excitations in this phase are obtained by acting with open strings on the ground state. The chapter goes on to review the theory of gapped boundaries and domain walls, with the associated module and bimodule category structures. These structures are then used to derive the most general tensor network representation of such a string-net ground state, a derivation which appears nowhere else in the literature. These tensor network states are characterised by the existence of matrix product operator (MPO) symmetries on the entanglement degrees of freedom, which play an important role in the remainder of this work.

In chapter 4, the tensor network constructions of the previous chapters are used to provide the representation theory for generalised symmetries in $(1+1)d$. This subsequently allows a classification of all possible duality transformations, non-trivial equivalences between models that map between different gapped phases. These duality transformations are realised as MPO intertwiners, which are mathematically described by module functors. These interact non-trivially with the boundary conditions of the theory, which can be studied systematically with the notion of bimodule tubes. In the second half of this chapter, these notions are generalised to $(2+1)d$, which essentially amounts to the categorification of the aforementioned results to fusion 2-categories, module 2-categories, and module 2-functors. Throughout, the various abstract notions are illustrated with representative examples, which greatly improve the readability of this chapter.

Chapter 5 contains the published lecture notes of a Les Houches lecture series on tensor networks. Like with everything in this thesis, these are meticulously crafted to a very high standard, and are quickly becoming a standard reference in the field. They collect a unique set of insights, and the explanations are clear with beautiful illustrations, which is a true testament to the writing and drawing skill of the author.

Chapter 6 concerns the classification of one-dimensional phases protected by a combination of translation and reflection symmetry, known as a frieze group, using tensor network technology. The relevant cohomology groups are computed, which requires going beyond the usual machinery because the reflection symmetry acts non-trivially on the underlying abelian group.

Chapter 7 connects the quantum information-theoretic definition of phases to the abstract categorical notion of Morita equivalence. More specifically, there is a redundancy in the Levin-Wen string-net construction based on a fusion category: distinct but Morita equivalent fusion categories produce the same topological phase. In this chapter, it is shown that this can be understood through the existence of a constant depth circuit that connects these two constructions. Particular care is taken with the excitations of the theory, yielding a map on the full Hilbert space, not just the ground state.

In chapter 8, it is shown that the classification of duality transformations argued in chapter 4 coincides with all possible ways of gauging the underlying symmetry. This establishes an important connection between two very powerful tools in quantum many body physics, and greatly facilitates the construction of gauge theories and duality transformations alike.

Many of the above results can be understood in the context of topological holography, which states that the symmetry content of a $(1+1)d$ theory can best be understood by realising it on the boundary of a $(2+1)d$ topologically ordered system. In chapter 9, it is shown that this logic can be reversed: one can construct a $(2+1)d$ topologically ordered state from a $(1+1)d$

theory as well through repeated gauging.

Finally, chapter 10 generalises the work on topological sectors in $(1+1)d$ reviewed in chapter 4 to the $(2+1)d$ case with abelian symmetries. This heavily uses the machinery of higher category theory, and is mathematically the most involved chapter in this thesis. Nevertheless, through worked examples, these abstract concepts are made concrete. The explicit realisation in terms of tensor network operators is expected to be of great importance in the construction of efficient numerical algorithms in $(2+1)d$.

APPRAISAL OF SCIENTIFIC VALUE

This thesis is a remarkable piece of work. The foundational chapters alone are of such quality that they could easily be published as a book, and I would urge the author to consider this. The work is characterised by a deep understanding of advanced mathematical concepts, made clear by the author's ability to translate them into explicit tensor network constructions that can be implemented on a computer. The study of generalised symmetries is an interdisciplinary field, requiring expertise in condensed matter physics, quantum information theory, and high energy physics. The author shows clear mastery in each of these subjects, and I expect these results to have profound impact on all of them. I have had the pleasure of working with the author at various stages of this PhD, and it is clear that he is an absolute expert in his field.

QUESTIONS AND REMARKS

The following are questions that arose during my reading of this thesis, and I look forward to discussing them during the private defence.

- The notion of (twisted) gauging can be used to exploit symmetries in variational tensor network algorithms. For example, a model in an SPT phase can be mapped to an equivalent one in a symmetry breaking phase, which is easier to simulate. Can something like this be done for an SPT phase protected by frieze symmetries?
- Regarding the iterated gauging construction of $(2+1)d$ topological phases, is there a physical understanding of why this works, and are there any ideas on how this could be used to realise general $(2+1)d$ topological phases?
- In $(1+1)d$, the underlying mathematical structure can also be understood algebraically through the theory of (weak) Hopf algebras. Is there an equivalent theory for the specific examples you consider in the $(2+1)d$ case?